

MA 574 : An Introduction to Probabilistic Machine Learning

Course Code: MA 574

Course Name: An Introduction to Probabilistic Machine Learning

L-T-P-C: 3-0-2-4

Students intended for: B. Tech/ MS/PhD/MSc

Elective or Core: Elective

Prerequisites: Probability and Statistics, Linear Algebra, and Optimizations

Mutual Exclusion: DS 413, CS 669

Approval: 59th BoA

Course Contents

- **Foundations of Probabilistic Reasoning:** Review of probability distributions and statistical inference. Maximum Likelihood Estimation (MLE), Bayesian inference, and Maximum a Posteriori (MAP) estimation. Fundamental information-theoretic concepts: entropy, conditional entropy, mutual information, and the Kullback-Leibler (KL) divergence. [6 Lectures]
- **Generative Models for Supervised Learning:** Generative approaches to classification: Naive Bayes and Gaussian Discriminant Analysis (GDA). Structured probabilistic models, Bayesian networks. Parameter estimation and inference techniques. [8 Lectures]
- **Probabilistic Discriminative Models:** Logistic regression and its theoretical foundation. Overfitting and generalization: regularization techniques. Introduction to the perceptron algorithm and comparison with probabilistic models. Overview of Bayesian logistic regression. [8 Lectures]
- **Ensemble Learning and Non-linear Extensions:** Non-linear and ensemble-based methods: decision trees, entropy, and information gain criteria. Ensemble methods: bagging, random forests, boosting, AdaBoost, and Gradient Boosting algorithms. Probabilistic interpretation of ensembles and the bias-variance trade-off. Introduction to Gaussian Processes (GPs) and how Bayesian non-parametric models define priors over functions. [10 Lectures]
- **Latent Variable Models and Unsupervised Learning:** Modeling unobserved structure: Expectation-Maximization (EM) algorithm, Gaussian Mixture Models (GMMs), and the method of moments. Unsupervised learning methods: k-means clustering and spectral clustering. Dimensionality reduction for high-dimensional data: probabilistic PCA and visualization techniques like t-SNE and UMAP. [10 Lectures]

Lab Class:

The lab component of this course focuses on the practical application of concepts via hands-on implementation with actual datasets. It aims to enhance students' understanding of concepts and strengthen their ability to use statistical and machine approaches and evaluate results effectively.

Text Books:

1. 1. Murphy, J. P., **Machine Learning: A Probabilistic Perspective**, The MIT Press, 2012.
2. 2. Mitchell, T. M., **Machine Learning**, McGraw Hill, 2017.



References:

1. Shai, S. S., and Shai, B. D., **Understanding Machine Learning: From Theory to Algorithms**, Cambridge University Press, 2014.
2. Duda, R. O., Hart, P. E., and Stork, D. G., **Pattern Classification**, 2nd Edition, Wiley-Interscience, 2000.
3. Tan, P. N., Steinbach, M., and Kumar, V., **Introduction to Data Mining**, Pearson, 2014.
4. Bishop, C. M., **Pattern Recognition and Machine Learning**, Springer, 2006.